SIMATS ENGINEERING

ASSESSMENT TOOL 2 – Concept Mapping

COURSE NAME: **Cryptography and Network Security**
COURSE CODE: **CSA 5103**

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 20%

QUESTIONS: 5 **Total Marks: 100**

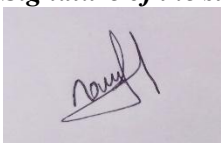
Annexure A

CONCEPT MAPPING

Code of Conduct:

I Naveen G 192411019 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Annexure A

CONCEPT MAPPING

1. Create a concept map linking the following: Security Attacks, Threats, Risks and Security Goals (Confidentiality, Integrity, Availability. Perform the following tasks:

   a. Show how each attack leads to specific threats and risks

   b. Map how security goals help mitigate them Provide one real-world example for each link

2. Develop a concept map comparing substitution techniques: Caesar Cipher, Monoalphabetic Cipher, Vigenère Cipher and Hill Cipher. Perform the following tasks to connect based on: Key usage, Complexity and Security strength. Analyze which technique is most secure and why.

3. Construct a concept map for Classical Encryption Schemes: Substitution Techniques and Transposition Techniques. Perform the following tasks:

   a. Show differences and similarities

      i. Map how each affects the Data structure and Security level

      ii. Analyze which is more vulnerable to attacks and why

4. Create a concept map connecting Steganography concepts: Data Hiding, Watermarking and Survivability. Perform the following tasks:

   a. Show relationships between these concepts

   b. Analyze how each contributes to secure communication

   c. Include one real-world application for each

5. Draw a concept map linking number theory concepts used in cryptography: Modular Inverse, Extended Euclid Algorithm, Fermat's Little Theorem, Euler's Phi Function and Euler's Theorem. Perform the following tasks:

   a. Show how each concept is mathematically connected

   b. Map their role in encryption/decryption

   c. Analyze which concept is most critical for public key cryptography and justify

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation

1) Security Attacks, Threats, Risks and Security Goals

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Security Attacks	Threats
Eavesdropping (Interception) Ex: Sniffing WiFi traffic to capture Login Credentials	Unauthorized disclosure of information
Data Tampering (modification) Ex: Modifying bank	Unauthorized modification of data
Masquerading (Impersonation) Ex: Attacker using Stolen Credentials to login	Unauthorized access
Denial of Service (DOS) Ex: DOS attack Sending huge traffic to a website	Resource exhaustion

Risks

Loss of Condfidentality

Sensitive data exposed

Loss of integrity Data

becames false or unreliable

unauthorized actions

in the System

Loss of availability

Security Goals (CIA)

Confidentiality

mitigation: Encryption

Access control

Integrity

mitigation: Hashing, digital

Signatures, Integrity checks

CIA Triad

mitigation: Strong Access

Authentication Control

Availability

mitigation: Firewalls Load

balancing Redundancy DOS

Protection

2) Substitution Technique

Each letter in the plain text is replaced by another letter or group of letters according to a rule or key

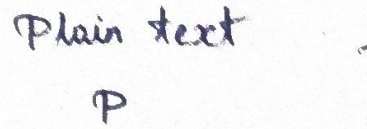
It is one of the simplest encryption technique

Substitution Techniques

Caesar Cipher	monoalphabetic cipher	Vigenere Cipher	Hill cipher

Parameter for Comparison

1) Key usage
2) Complexity
3) Security Strength



General mode

Caesar Cipher

uses a Signal Shift value

Same Shift for all letters

$C = P + K \pmod{26}$

Vigenere Cipher

uses a keyword (multiple Shift)

Repeating keywords decides the shift for each position

more complex than Caesar and monoalphabetic

monoalphabetic Cipher

uses a fixed substitution alphabet

Each letter is replaced with letter

one to one mapping

Hill Cipher

Uses a matrix as key

Substitutes blocks of letters

use matrix multiplication

Security Analysis

Caesar Cipher : only 25 possible keys Easily broken by brute force

Monoalphabetic Cipher : Can be broken using frequency analysis
Same letter mapped to same letter

Vigenere Cipher : Harder to break than Caesar and monoalphatic
Resists simple frequency analysis

Hil cipher : most secure among the four. use matrix key and substitution blocks of letter making it more resistant to frequency analysis